

Electrons do not move freely around a nucleus, but rather are confined to regions of space called **principal energy levels (shells)**.

❑ **Shells** are numbered as **1, 2, 3** from the inside out.

❑ Each shell can contain up to **$2n^2$** electrons (**n**= number of shell)

TABLE 1.1 Distribution of Electrons in Shells		
Shell	Number of Electrons Shell Can Hold	Relative Energies of Electrons in Each Shell
4	32	
3	18	
2	8	
1	2	

- Shells are divided into **subshells** designated by the letters *s*, *p*, *d* and *f*
- Within these subshells, electrons are grouped in **orbitals**
- Orbital** represents the space where an electron spends most (90-95%) of its time

TABLE 1.2 Distribution of Orbitals within Shells		
Shell	Orbitals Contained in Each Shell	Maximum Number of Electrons Shell Can Hold
4	One 4 <i>s</i> , three 4 <i>p</i> , five 4 <i>d</i> , and seven 4 <i>f</i> orbitals	$2 + 6 + 10 + 14 = 32$
3	One 3 <i>s</i> , three 3 <i>p</i> , and five 3 <i>d</i> orbitals	$2 + 6 + 10 = 18$
2	One 2 <i>s</i> and three 2 <i>p</i> orbitals	$2 + 6 = 8$
1	One 1 <i>s</i> orbital	2

- The 1st shell ($n = 1$) has only an *s* orbital (1*s*)
- The 2nd shell ($n = 2$) has *s* and *p* orbitals (2*s* and 2*p*)
- The 3rd shell ($n = 3$) has *s*, *p*, and *d* orbitals (3*s*, 3*p*, and 3*d*)

Quantum numbers

There are 4 identification numbers called quantum numbers which can fully describe an electron in an atom.

1. Principal quantum number, n :

This quantum number denotes the principal shell to which the electrons belongs.

Principal quantum number ($n =$)	1	2	3	4
Letter designation	K	L	M	N
Maximum number of electrons ($2n^2 =$)	2	8	18	32

2. Azimuthal quantum number, l (also called Angular momentum quantum number):

- The quantum number l defines the shape of the orbital occupied by the electron and the angular momentum of the electron.
- For any given value of n , the l may have integer values from 0 to $n-1$, each of which refers to an *Energy sublevel or Sub-shell*.
- These *sublevels* are symbolised by letters s, p, d, f .

For example, for principal quantum number $n = 1$, the only possible value for l is 0 *i.e.*, there is only one possible subshell *i.e.* s-subshell ($n = 1, l = 0$). For $n = 2$, there are two possible values of l , $l = 0$ and $l = 2 - 1 = 1$.

$$l = n-1$$

The different **subshells** are **denoted by letters** as follows:

Letter	s	p	d	f	g
l	0	1	2	3	4

$n=1$

$l=0 (1s)$

$n=2$

$l=0 (2s)$

$l=1 (2p)$

$n=3$

$l=0 (3s)$

$l=1 (3p)$

$l=1 (3d)$

$n=4$

$l=0 (4s)$

$l=1 (4p)$

$l=2 (4d)$

$l=3 (4f)$

$n=5$

$l=0 (5s)$

$l=1 (5p)$

$l=2 (5d)$

$l=3 (5f)$

$l=4 (5g)$

3. Magnetic quantum number, m_l

- Magnetic quantum number (m_l) describes the orientation of the orbital in space.
- Within a subshell, the value of m_l depends on the value of the angular momentum quantum number, l
- For a certain value of l , there are $(2l + 1)$ integral values of m_l as follows:
 $-l, (-l + 1), \dots, 0, \dots, (+l - 1), +l$

- If $l = 0$, then $m_l = 0$
- If $l = 1$, then there are $[(2 \times 1) + 1] = 3$ values of m_l .

These are: -1, 0, and 1

- If $l = 2$, then there are $[(2 \times 2) + 1] = 5$ values of m_l .

These are: -2, -1, 0, +1 and +2.

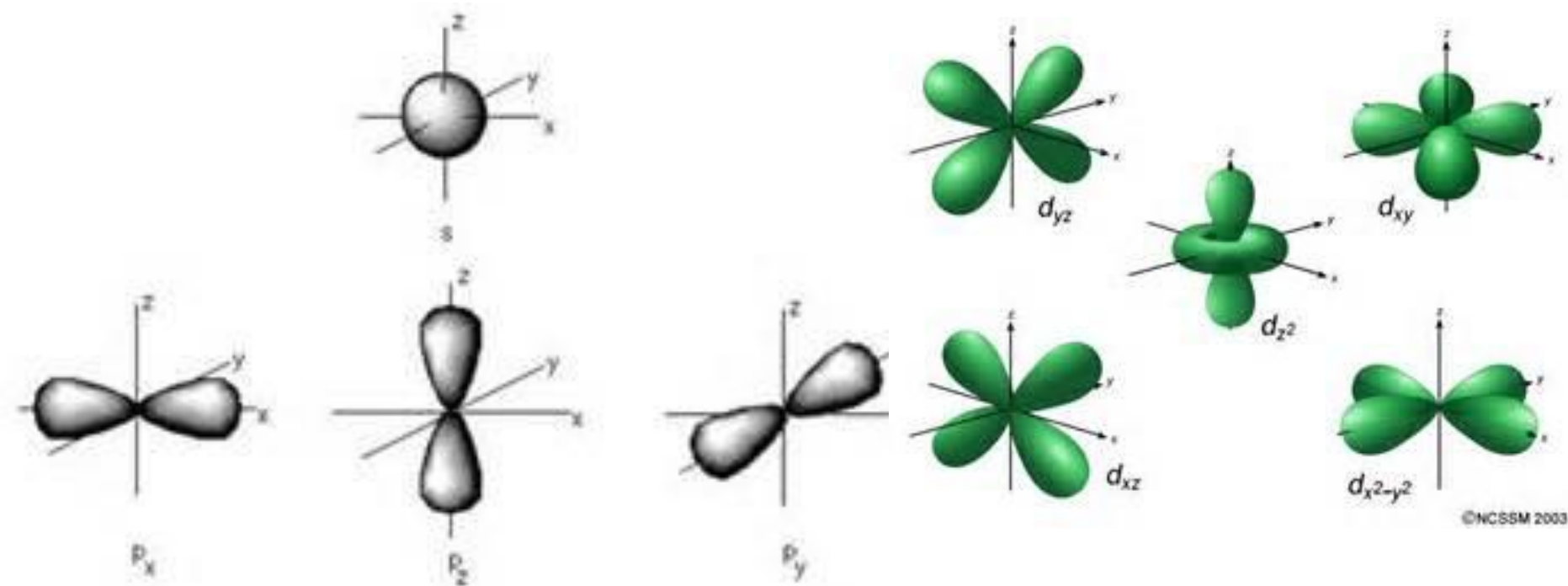
4. Spin quantum number, m_s

This quantum number refers to the **two possible orientations of the spin axis** of an electron; possible values are: $+1/2$ and $-1/2$.

TABLE 2.1. QUANTUM NUMBERS AND ELECTRON ACCOMMODATION

Principal Q-number n		Azimuthal Q-number l		Magnetic Quantum Number m	Spin Quantum Number s	Number of Electrons accommodated
1	K	0	s	0	$+\frac{1}{2}, -\frac{1}{2}$	2
2	L	0	s	0	$+\frac{1}{2}, -\frac{1}{2}$	2
		1	p	+1, 0, -1	$+\frac{1}{2}, -\frac{1}{2}$	6
3	M	0	s	0	$+\frac{1}{2}, -\frac{1}{2}$	2
		1	p	+1, 0, -1	$+\frac{1}{2}, -\frac{1}{2}$	6
		2	d	+2, +1, 0, -1, -2,	$+\frac{1}{2}, -\frac{1}{2}$	10
4	N	0	s	0	$+\frac{1}{2}, -\frac{1}{2}$	2
		1	p	+1, 0, -1	$+\frac{1}{2}, -\frac{1}{2}$	6
		2	d	+2, +1, 0, -1 -2	$+\frac{1}{2}, -\frac{1}{2}$	10
		3	f	+3, +2, +1, 0, -1, -2, -3	$+\frac{1}{2}, -\frac{1}{2}$	14

Atomic Orbitals



- **S Atomic orbital** is spherically symmetrical
- **P Atomic orbitals** are not spherically symmetrical. They are dumbbell-shaped
- **d Atomic orbitals** are cloverleaf-shaped

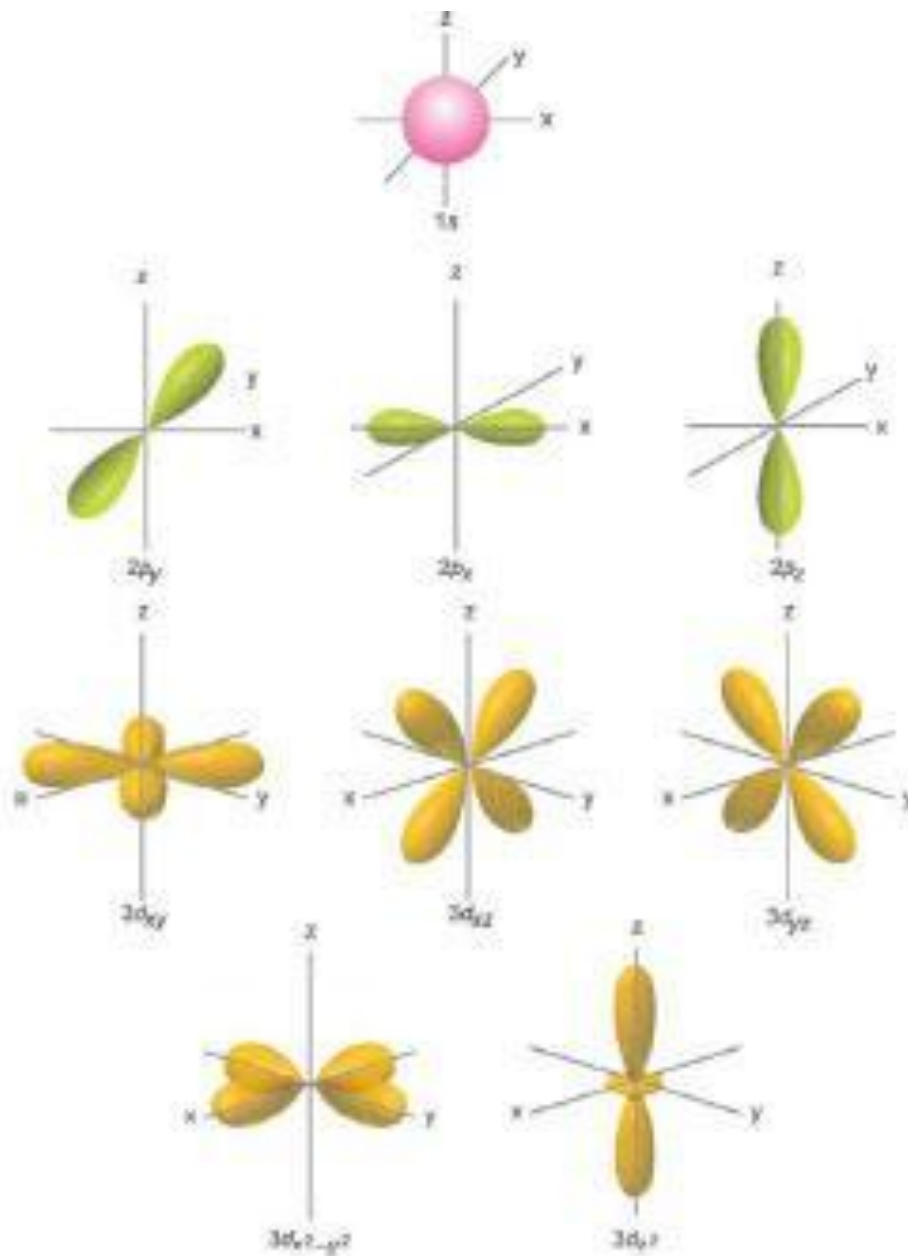


Figure 2.8
Shapes and orientation of s, p and d orbitals.

Electron Configuration of Atoms

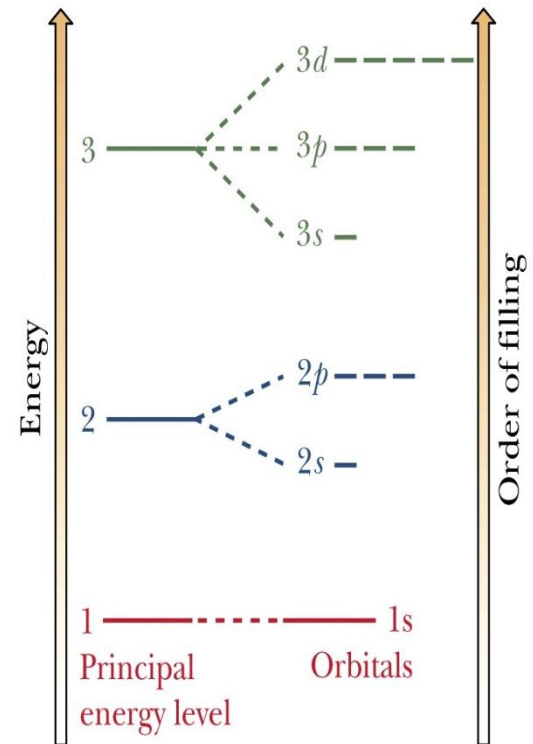
The electron configuration of an atom is a description of the orbitals the electrons in the atom occupy.

Ground state electron configuration:

Three rules:

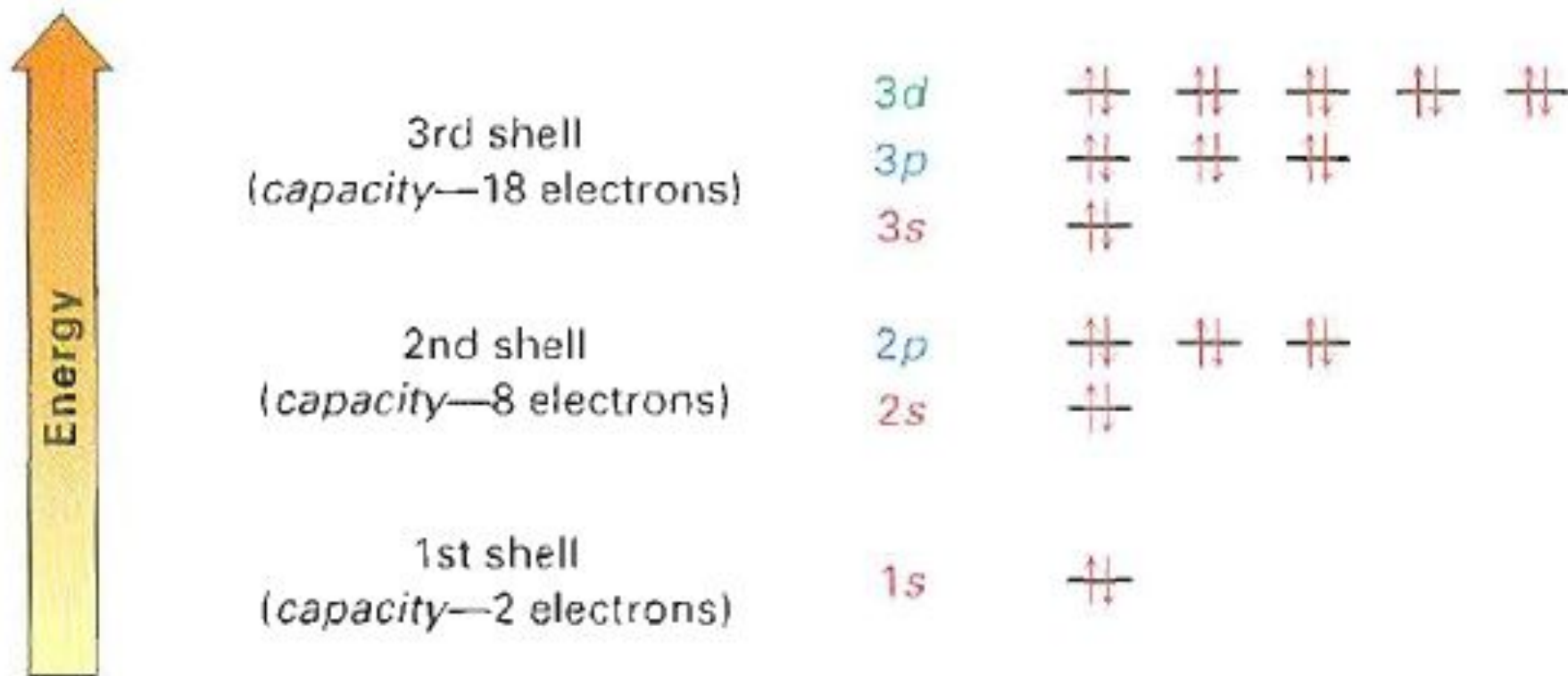
Rule 1: Orbitals fill in order of increasing energy from lowest to highest.

Rule 2: Each orbital can hold up to two electrons with their spins paired.

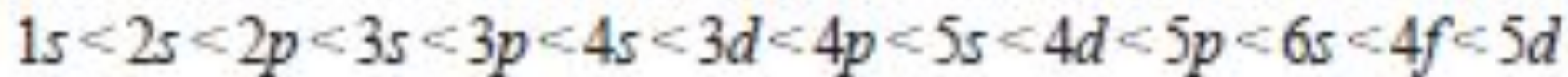


Rule 3: For orbitals with equivalent energy, add one electron to each equivalent orbitals before a second electron is added to any one of them.

Electron Configuration of Atoms



The increasing order of energy of various orbitals is as follows:



Ground-state Electron Configurations of Some Elements

TABLE 1.3 Ground-State Electron Configurations for Elements 1–18

First Period*	Second Period	Third Period
H 1 $1s^1$	Li 3 [He] $2s^1$	Na 11 [Ne] $3s^1$
He 2 $1s^2$	Be 4 [He] $2s^2$	Mg 12 [Ne] $3s^2$
	B 5 [He] $2s^2 2p_x^1$	Al 13 [Ne] $3s^2 3p_x^1$
	C 6 [He] $2s^2 2p_x^1 2p_y^1$	Si 14 [Ne] $3s^2 3p_x^1 3p_y^1$
	N 7 [He] $2s^2 2p_x^1 2p_y^1 2p_z^1$	P 15 [Ne] $3s^2 3p_x^1 3p_y^1 3p_z^1$
	O 8 [He] $2s^2 2p_x^2 2p_y^1 2p_z^1$	S 16 [Ne] $3s^2 3p_x^2 3p_y^1 3p_z^1$
	F 9 [He] $2s^2 2p_x^2 2p_y^2 2p_z^1$	Cl 17 [Ne] $3s^2 3p_x^2 3p_y^2 3p_z^1$
	Ne 10 [He] $2s^2 2p_x^2 2p_y^2 2p_z^2$	Ar 18 [Ne] $3s^2 3p_x^2 3p_y^2 3p_z^2$

*Elements are listed by symbol, atomic number, and ground-state electron configuration, in that order.

TABLE 2.3. GROUND STATE ELECTRON CONFIGURATION OF ELEMENTS					
Z	Element	Electron configuration	Z	Element	Electron configuration
1	H	$1s^1$	21	Sc	$[\text{Ar}]3d^14s^2$
2	He	$1s^2$	22	Ti	$[\text{Ar}]3d^24s^2$
3	Li	$[\text{He}]2s^1$	23	V	$[\text{Ar}]3d^34s^2$
4	Be	$[\text{He}]2s^2$	24	Cr	$[\text{Ar}]3d^54s^1$
5	B	$[\text{He}]2s^22p^1$	25	Mn	$[\text{Ar}]3d^54s^2$
6	C	$[\text{He}]2s^22p^2$	26	Fe	$[\text{Ar}]3d^64s^2$
7	N	$[\text{He}]2s^22p^3$	27	Co	$[\text{Ar}]3d^74s^2$
8	O	$[\text{He}]2s^22p^4$	28	Ni	$[\text{Ar}]3d^84s^2$
9	F	$[\text{He}]2s^22p^5$	29	Cu	$[\text{Ar}]3d^{10}4s^1$
10	Ne	$[\text{He}]2s^22p^6$	30	Zn	$[\text{Ar}]3d^{10}4s^2$
11	Na	$[\text{Ne}]3s^1$	31	Ga	$[\text{Ar}]3d^{10}4s^24p^1$
12	Mg	$[\text{Ne}]3s^2$	32	Ge	$[\text{Ar}]3d^{10}4s^24p^2$
13	Al	$[\text{Ne}]3s^23p^1$	33	As	$[\text{Ar}]3d^{10}4s^24p^3$
14	Si	$[\text{Ne}]3s^23p^2$	34	Se	$[\text{Ar}]3d^{10}4s^24p^4$
15	P	$[\text{Ne}]3s^23p^3$	35	Br	$[\text{Ar}]3d^{10}4s^24p^5$
16	S	$[\text{Ne}]3s^23p^4$	36	Kr	$[\text{Ar}]3d^{10}4s^24p^6$
17	Cl	$[\text{Ne}]3s^23p^5$	37	Rb	$[\text{Kr}]5s^1$
18	Ar	$[\text{Ne}]3s^23p^6$	38	Sr	$[\text{Kr}]5s^2$
19	K	$[\text{Ar}]4s^1$	39	Y	$[\text{Kr}]4d^15s^2$
20	Ca	$[\text{Ar}]4s^2$	40	Zr	$[\text{Kr}]4d^25s^2$

TABLE 2.2. ANOMALOUS ELECTRON CONFIGURATION OF SOME ELEMENTS

Element	At. No.	Expected Configuration	Actual Configuration
Cr	24	$4d^4 4s^2$	$3d^5 4s^1$
Cu	29	$3d^9 4s^2$	$3d^{10} 4s^1$
Mo	42	$4d^4 5s^2$	$4d^5 5s^1$
Pd	46	$4d^8 5s^2$	$4d^{10} 5s^0$
Ag	47	$4d^9 5s^2$	$4d^{10} 5s^1$
Pt	78	$5d^8 6s^2$	$5d^9 6s^1$
Au	79	$5d^9 6s^2$	$5d^{10} 6s^1$